

AIoT UPPER-ARM WEARABLE

# Nova-Band

An AIoT-driven upper-arm wearable for real-time physiological monitoring and running performance

## TEAM NOVA-BAND

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PPG + IMU

EDGE FFT · ESP32

EARLY WARNING

IN COLLABORATION WITH



UPPER-ARM FIT · PPG + IMU · AIoT

# What we will cover

Five chapters, from the running-injury problem to evidence and market.

**01**

## The running problem

Sedentary lifestyles, overtraining and unreliable wrist data.

**02**

## The Nova-Band system

Upper-arm fit, PPG + IMU, edge FFT and an AI early warning.

**03**

## Product & hardware

Module anatomy, 3D digital twin and specifications.

**04**

## Software & AI

The runner app, Readiness Score and the community dashboard.

**05**

## Evidence & market

MAPE and SUS results, market size and projection.

# Background

Physical inactivity is a public-health problem — and running is where many people start.



## 37.4%

of Indonesians aged  $\geq 10$  are not active enough

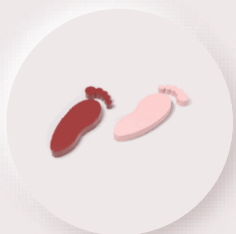
Below the recommended level of physical activity. SKI 2023



## 3 Million+

deaths a year linked to physical inactivity

One of the leading global risk factors for premature death. WHO 2020



## 1–42%

MAPE of consumer wearables in step counting

Accuracy varies widely between devices. Al-Marroof et al. 2024



## 15–44%

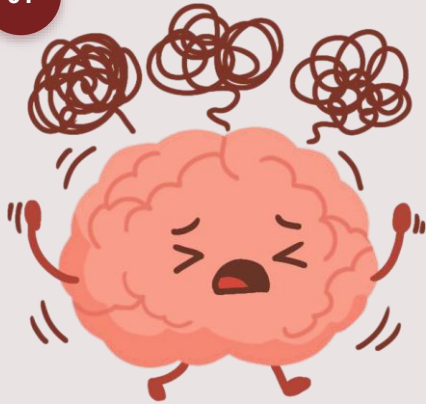
MAPE in energy / calorie estimation

Even wider error on calories burned. Al-Marroof et al. 2024

# Meet the **problem**

Four ways today's running culture — and wrist wearables — fail runners.

01



## Worried-well anxiety

Excessive anxiety about digital metrics on the watch screen.

02



## Physical harm

Runners ignore pain and extreme fatigue to hit on-screen targets.

03



## Overtraining risk

Pushing beyond physical limits, driven by FoMO and lifestyle pressure.

04

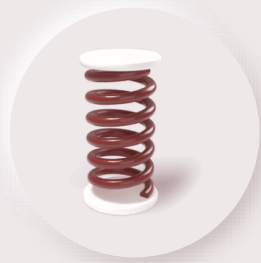


## Data inaccuracy

Wrist sensors lose accuracy exactly when running gets intense.

# Trends in runner injuries

The injury pattern has shifted toward heavier soft-tissue and tendon damage.



## Running is booming — so is overtraining

Among urban runners, lifestyle pressure, commodification and Fear of Missing Out push people past their physical limits. Injuries used to be dominated by kneecap pain; they now shift to heavier soft tissue and tendons.

**Two main predictors** excess mileage · previous injury history

1

### Achilles tendinopathy

Tendon overload behind the ankle

2

### Calf strain

Gastrocnemius injury under repeated push-off

3

### Hamstring & quadriceps strain

Thigh muscles overstretched without adequate warm-up

4

### Iliotibial Band Syndrome (ITBS)

Outer-knee pain linked to high training load

5

### Knee meniscus tear

Cartilage damage from accumulated impact

# Research questions

Two problems formulated, two objectives set.

## PROBLEM FORMULATION

1

How do motion artifacts from arm swing affect the accuracy of heart-rate and motion data in wrist-worn wearables used by runners?



2

How can Nova-Band be designed and tested to extract an accurate Readiness Score that helps prevent soft-tissue injury and cardiovascular collapse?



## RESEARCH OBJECTIVES

1

Design Nova-Band as an upper-arm wearable that mechanically stabilises the sensors to minimise motion artifacts and produce accurate PPG and IMU data.



2

Evaluate its effectiveness through MAPE accuracy testing and SUS usability testing in extracting a reliable Readiness Score.



# State of the **art**

What the literature shows — and the gap Nova-Band closes.

**Siradj**  
2016

Wearables help runners track physical activity practically and independently for self-motivation.



**Gap** Consumer accuracy is low: MAPE 1–42% for steps, 15–44% for calories.

**Al-Maroofo et al.**  
2024

Smartwatches enable continuous tracking of physiological metrics and self-monitoring interventions.



**Gap** Reliability and validity vary across devices, especially at high intensity.

**Moghaddam et al.**  
2025

Placement studies show wrist-worn devices are practical for everyday heart-rate monitoring.



**Gap** Wrist PPG is highly vulnerable to motion artifacts during intense running.

**McGrath & Doyle et al.**  
2023

IMU sensors with machine learning estimate ground reaction force and running kinematics precisely.



**Gap** Largely lab-grade — not yet integrated for real-time daily use by runners.



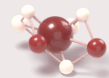
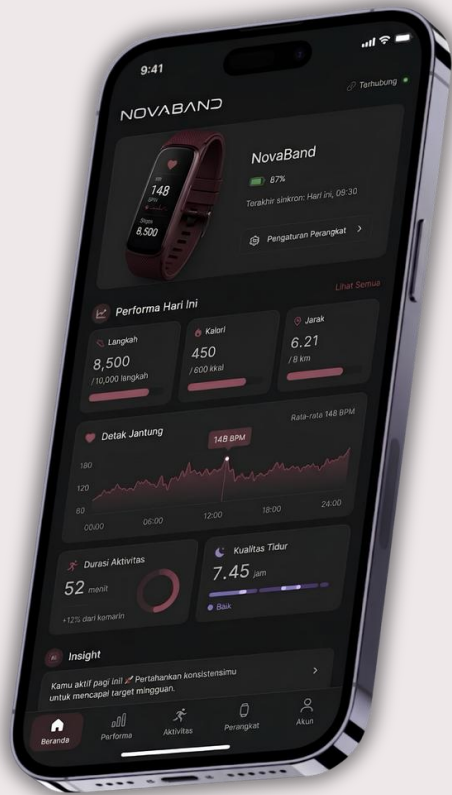
**Nova-Band's answer** ✓ Upper-arm placement reduces motion artifacts

✓ PPG, IMU and ESP32 integrated: cardio + biomechanics

✓ IoT dashboard alerts organisers early to prevent overtraining

# The solution

Four layers of protection in one band.



## AI & IoT Synergy

Sensor readings from the band (IoT) are processed continuously by AI into physiological insight.



## Shared Safety Net

A dual-layer system: alerts reach the runner and the supervisor at the same time to prevent collapse.



## Biological Early Warning

Proactive, instant alerts the moment a runner starts pushing past the body's safe limits.



## Soft Network Load

IMU kinematic tracking of mechanical stress on muscles and tendons — against Achilles tendinopathy & ITBS.

# Our novelty

Three things a wrist wearable does not do together.

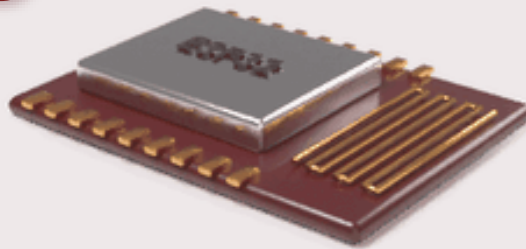
1



## Upper-arm fit

Worn away from the arm's largest swing, so the optics stay pressed to the skin and motion artifacts drop — a mechanical fix first.

2



## Edge FFT on the ESP32

A Fast Fourier Transform on the device separates heart-rate frequencies from cadence noise before any data leaves the band.

3



## Shared safety net

Data syncs to a community dashboard, so an event supervisor sees an alert at the same moment the runner does.

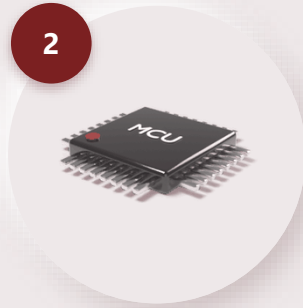
# How Nova-Band **protects** runners

From a beam of light on the skin to an alert on the supervisor's screen.



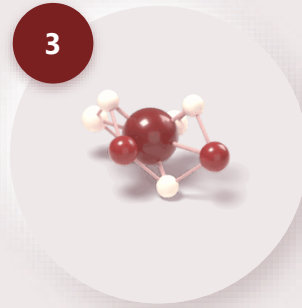
## Acquire

PPG and IMU sampled together on the upper arm.



## Filter

FFT on the ESP32 strips cadence noise from the signal.



## Analyse

AI turns clean signals into a Readiness Score.



## Notify

Bluetooth streams metrics to the runner's app.



## Protect

The IoT dashboard watches the whole group.



### An alert reaches two people at once

When a runner crosses a safe limit, the warning goes to the runner and to the event supervisor simultaneously — the Shared Safety Net.

# Hardware operation (PPG)

Photoplethysmography: every heartbeat changes how much light the skin gives back.



## LED Sensor

Green LEDs shine into the skin of the upper arm.



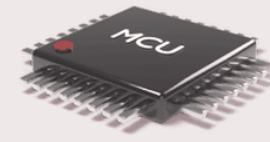
## Blood Absorbs Light

Blood volume changes with each beat, absorbing light.



## Photodetector

Captures the light reflected back from the skin.



## Microcontroller

The ESP32 filters the signal with an on-device FFT.



## Heart Beats

Becomes real-time BPM and feeds the Readiness Score.

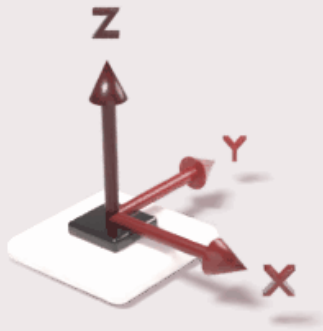
## What the band measures

An LED shines into the skin; with every heartbeat the blood volume below changes, and so does the light that returns to the photodetector. The ESP32 filters that signal and turns it into heart rate.

**2.5–3.5%** MAPE

# Hardware operation (IMU)

Three motion sensors that watch for emergencies.



## Accelerometer

### FALLS & COLLAPSE

Detects drastic changes in motion speed to identify sudden falls or physical collapse.



## Gyroscope

### POSTURE & BALANCE

Monitors body posture and tilt to detect loss of balance, dizziness or unsteady movement.



## Magnetometer

### ORIENTATION

Determines precise body orientation to assist spatial tracking in emergency situations.

# Meet our **product**

An upper-arm band that keeps its sensors still while you run.



## Upper-arm fit

Sits away from the arm's largest swing — the optics stay in contact at running pace.



## PPG + IMU in one module

Heart rate and body movement captured together, filtered on the ESP32.



## Built to be worn

Precision 3D-printed, sweat-resistant casing on a porous modular elastic strap.



NOVA-BAND · PRODUCT

# Product anatomy

Five layers inside the module, from the optics to the skin.



Blender model · internal layout illustrative

1

## Optical window

Two green LEDs, an IR diode and a photodetector

2

## Casing

Precision 3D-printed polymer, sweat-resistant

3

## Li-Po battery

High-density cell for long endurance runs

4

## Mini PCB

ESP32, IMU and step-down regulator

5

## Sensor pod

Skin side — keeps the optics in contact

# 3D digital twin

A Blender model rebuilt from the product images — and reused everywhere.



PRODUCT IMAGE



BLENDER 3D

## One model, every medium

- ✓ Interactive 3D viewer on the website
- ✓ 46-asset icon & render library
- ✓ Animated loops in this deck
- ✓ 1920 × 1080 showreel video

Modelled from the product images: curved pill module, framed optical window, woven strap with stitched keeper.

# Upper-arm fit

Why the band lives on the upper arm, and how it is used.



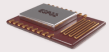
## Why the upper arm?

The wrist swings the most while running, and that swing bleeds into the optical signal. The upper arm moves less, so the sensor stays in contact.

- 1 Strap the band on the upper arm, module facing out.
- 2 Pair it with the Nova-Band app over Bluetooth.
- 3 Run — heart rate, cadence and load stream in real time.
- 4 Alerts reach you and your supervisor if a limit is crossed.

# Specifications

What is inside the prototype.



## MICROCONTROLLER

**ESP32 on a custom mini PCB**



## MOTION SENSOR

**IMU — accelerometer, gyroscope, magnetometer**



## CASING

**Precision 3D-printed polymer**



## POWER

**Li-Po + step-down regulator**



## PHYSIOLOGICAL SENSOR

**Optical PPG**



## PROCESSING

**On-device FFT, C++ firmware**



## STRAP

**Porous modular elastic**



## CONNECTIVITY

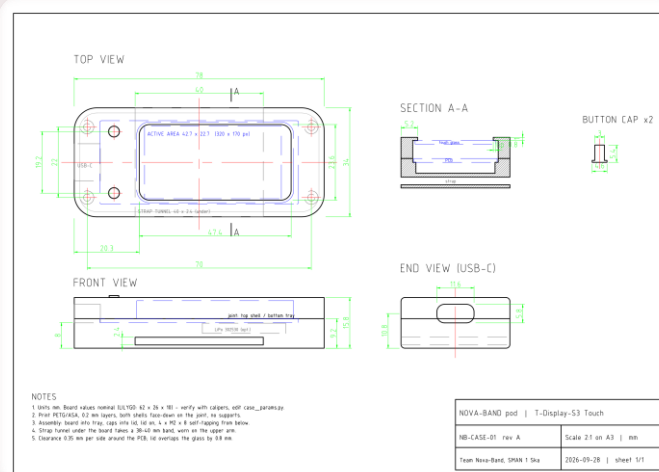
**Bluetooth → app → IoT dashboard**

# Working prototype

A LILYGO T-Display-S3 Touch in a 3D-printed pod — sketched in LibreCAD, solid-modelled in FreeCAD.



PRINTED POD · PETG



LibreCAD sketch · NB-CASE-01

Top, front, end and section views, A3 at 2:1. Board values are nominal — measured before printing.

DXF · STEP · STL

## MCU

ESP32-S3R8 · 240 MHz dual core

## MEMORY

16 MB flash · 8 MB PSRAM

## DISPLAY

1.9" IPS 320×170 · touch

## CASE

78 × 34 × 15.8 mm · 2 parts + caps

## STRAP

38–40 mm band, under-tunnel

## LINK

Bluetooth LE · USB-C

# On-device interface

Firmware for the band's own screen — Blender assets, depth of field, 60 fps on both CPU cores.



## Blender splash

A rack-focus intro rendered in Blender, played from flash.



## Depth of field

Gym bokeh far, glass cards mid, blurred specks near — three parallax layers.



## 60 frames per second

Each core draws half the frame; DMA sends it to the ST7789.



## Touch & buttons

Swipe with spring physics; long-press to start or pause a run.

# Connected to the **app**

Bluetooth LE or USB, one JSON protocol — and the web app installs like a native app (PWA).



NovaBand · pairing page

## ⇄ BLUETOOTH LE

Web Bluetooth · Android, Windows, macOS

## ⇄ USB-C

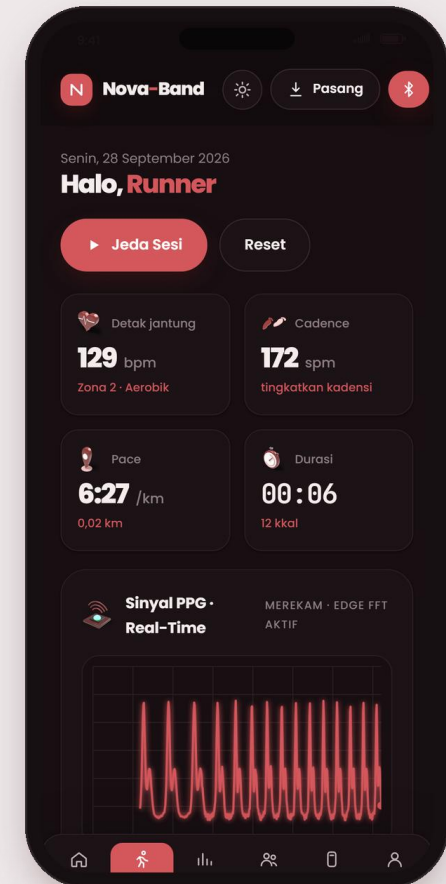
Web Serial · desktop Chrome / Edge

Telemetry 1 Hz · PPG wave 50 Hz

Standard Heart Rate service 0x180D

App → band: run, alerts, coach messages

The QR code on the band opens **novaband-id.web.app** — installable, works offline.



# In the gym



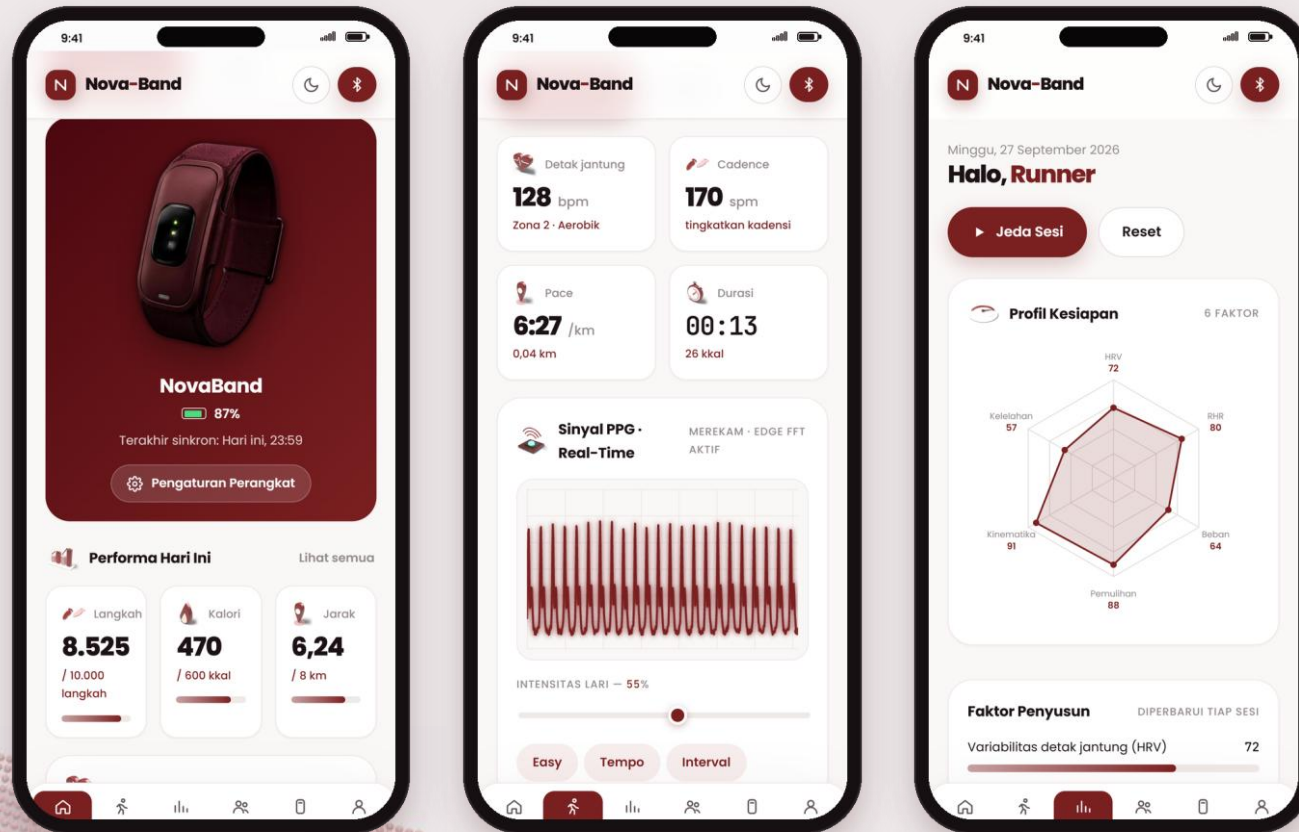
## Rendered, not photographed

Blender Cycles, from the CLI. The gym, runner and props are Sketchfab models (CC-BY); the pod, its live screen and the band are ours.



# Smart application

The runner app that ships with the band — real-time and explainable.



## Today's performance

Steps, calories and distance against daily goals.



## Live PPG signal

Filtered on the band and streamed while you run.



## Readiness profile

Six factors behind one daily score.



## Early warning

Alerts when a safe limit is crossed.

# Readiness score & **early warning**

Physiology and movement, read together — so every run stays inside safe limits.



## Read

Heart rate from PPG; cadence and load from the IMU.



## Analyse

Six factors combined into one Readiness Score.



## Warn

Alert runner and supervisor when a limit is crossed.



# 85

Ready to train

### THE SIX FACTORS · DEMO APP

Heart-rate variability

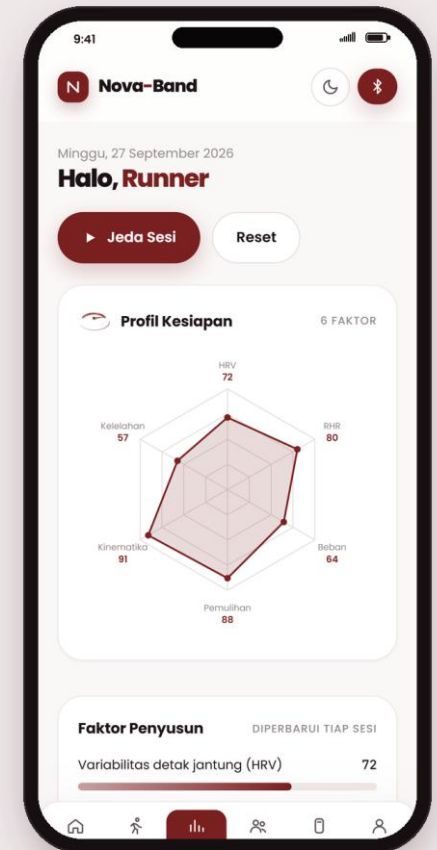
Resting heart rate

Acute : chronic load

Recovery quality

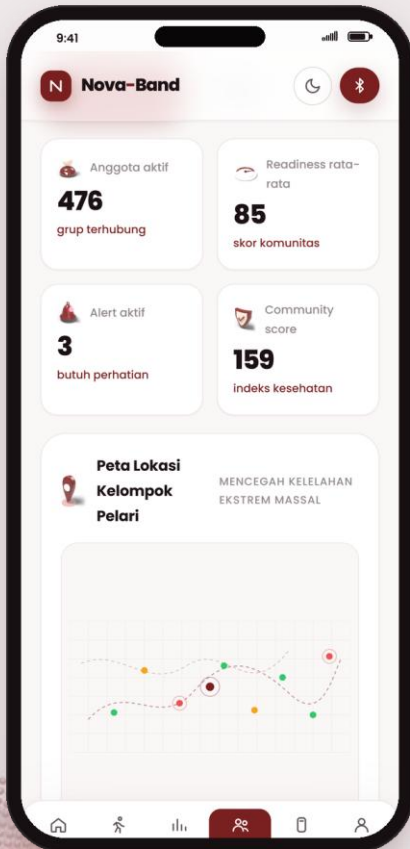
Kinematic consistency

Accumulated fatigue



# Community dashboard

The Shared Safety Net for event supervisors and running groups.



## Group map

Every runner's position and status on one map.



## Member status

Heart rate, readiness and distance for each runner.



## Simultaneous alerts

The supervisor is notified at the same moment as the runner.

# 476

runners connected

demo data

# 3

active alerts

demo data

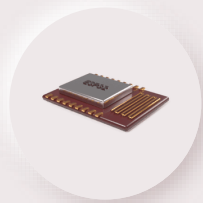
# Method & timeline

Research & Development with a quantitative experimental approach.

**1**

## *Design Stage*

- Ergonomic casing designed and 3D-printed
- Upper-arm placement to damp shocks
- PPG, IMU and ESP32 selected

**2**

## *Integration Stage*

- Mini PCB, Li-Po and step-down assembled
- Sensor fusion + FFT programmed in C++
- Bluetooth link to app and IoT dashboard

**3**

## *Validation Stage*

- MAPE vs high-speed motion capture
- SUS usability test, 60 respondents
- Findings feed the next design iteration

R&D · quantitative experimental

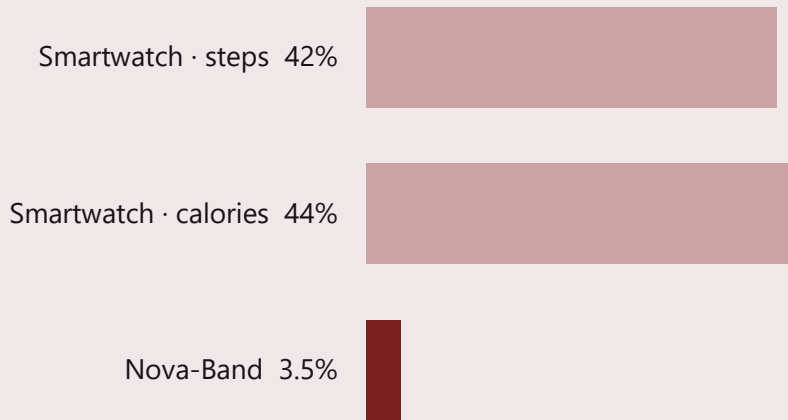
MAPE vs high-speed motion capture

SUS · 60 respondents

# Result & discussion

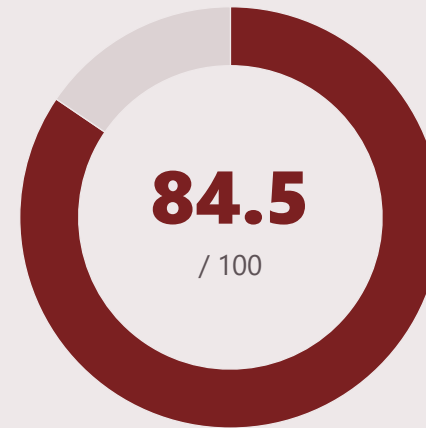
Accurate in the field, comfortable to wear.

## Hardware accuracy · MAPE (%)



Nova-Band: 2.5–3.5% — below the 5% limit for professional sports instruments. Consumer values are the upper bounds of reported ranges.

## Usability · SUS



### Grade A · Acceptable

60 respondents · 10 SUS statements · above the 68 threshold



TESTING DOCUMENTATION

# Market opportunity

Running events are where group safety matters most.

**TAM**

**SAM**



**1,818**

**TAM · Indonesia**

running events held nationwide in 2025



**302**

**SAM · Central Java**

running events in the province



**18**

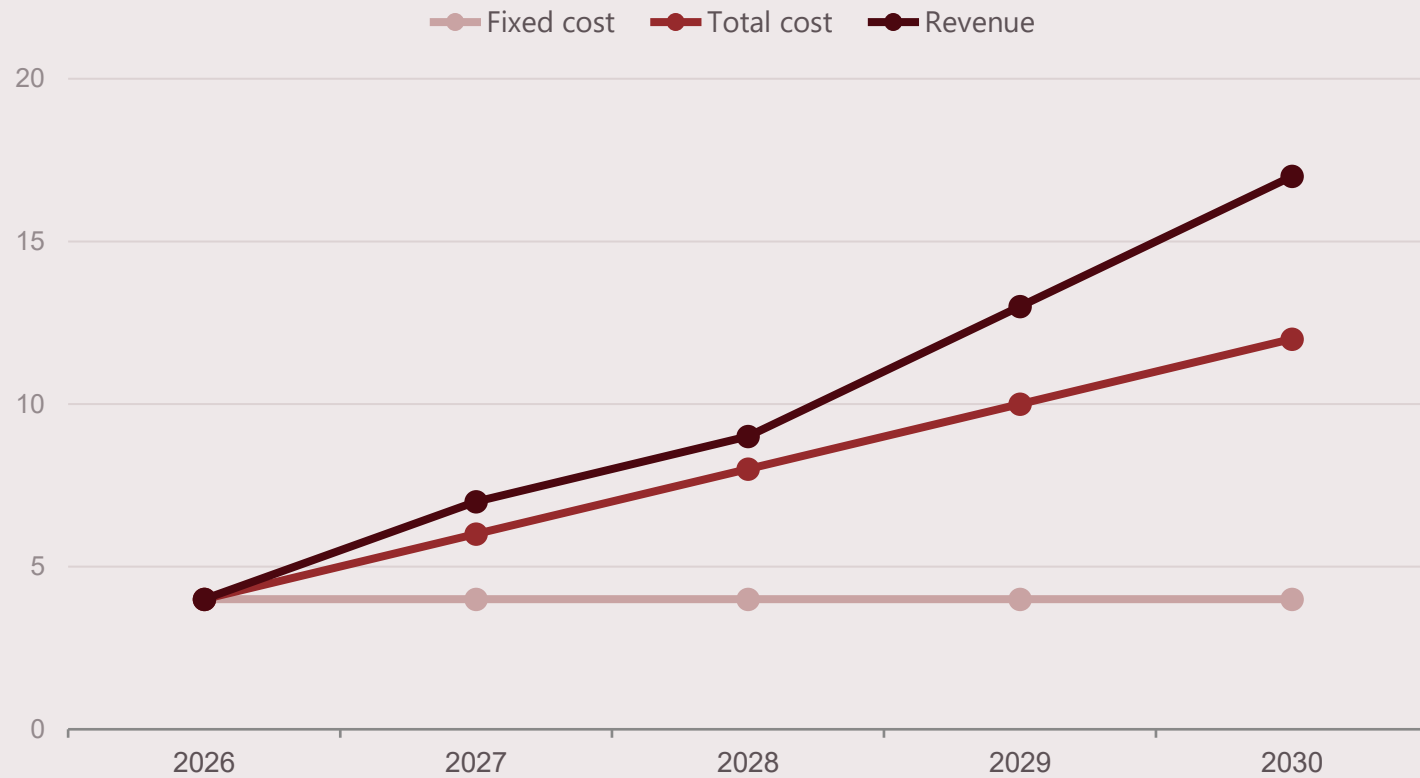
**SOM · Surakarta**

running events across the city

Source: 2025 running-event data, verified by the team.

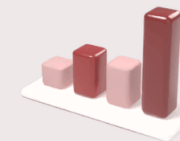
# Business projection

Fixed cost, total cost and revenue, 2026–2030.



## Revenue above total cost from 2027

The gap widens every year, to 17 vs 12 by 2030.



## Fixed cost held flat

Values read from the chart in the ISIF deck; units as in the source.

# Our partnership

Collaborators supporting research, validation and testing.



**REKA Mersif**

Collaborator



**SMA Negeri 1 Surakarta**

Research team



**Universitas Sebelas Maret**

Collaborator



**Mersif Academy · Enuma  
Technology**

Collaborators



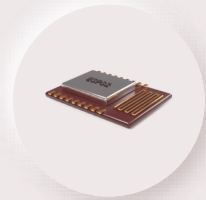
**SMA Negeri 5 Yogyakarta**

ISIF 2026 partner school



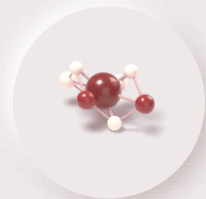
**Partnership · ISIF 2026** Built by students of SMA Negeri 1 Surakarta together with university, industry and school partners.

# Conclusion and **copyright**



## System integration

PPG (heart rate) and IMU (body movement) on an ESP32, worn on the upper arm, collect physiological and motion data simultaneously.



## AI analysis

AI processes activity patterns and physiological responses together, in real time, during training.



## Impact & final result

An accurate, practical, ergonomic and data-driven wearable for real-time runner performance monitoring.

## COPYRIGHT & IP



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SMA Negeri 1 Surakarta

# 3D asset credits

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# Thank you

*Monitor. Analyze. Improve.*

Real-time physiological monitoring for safer running — one band, one community at a time.

## TEAM NOVA-BAND

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